

User Manual

for S32G CAN_LLCE Driver

Document Number: UM11CAN_LLCEASR4.4 Rev0000R1.0.7 Rev. 1.0

1 Revision History	2
2 Introduction	3
2.1 Supported Derivatives	3
2.2 Overview	4
2.3 About This Manual	4
2.4 Acronyms and Definitions	5
2.5 Reference List	5
3 Driver	6
3.1 Requirements	6
3.2 Driver Design Summary	6
3.3 Hardware resources	7
3.4 Deviation from Requirements	7
3.4.1 LLCE Headless Mode	15
3.5 Driver Limitations	15
3.6 Driver Usage and Configuration Tips	16
3.6.1 Driver APIs usage	16
3.6.2 Can Hardware Object Handle (HOH) configuration	17
3.6.3 Configuration Tips	18
3.6.4 LLCE Headless Mode	18
3.7 Runtime Errors	18
3.8 Symbolic Names Disclaimer	19
4 Tresos Configuration Plug-in	20
4.1 Module Can	24
4.2 Container CanGeneral	25
4.3 Parameter CanDevErrorDetect	25
4.4 Parameter CanEnableUserModeSupport	25
4.5 Parameter CanMulticoreSupport	26
4.6 Parameter HostInterface	26
4.7 Parameter CanVersionInfoApi	27
4.8 Parameter CanIndex	27
4.9 Parameter CanMainFunctionBusoffPeriod	28
4.10 Parameter CanMainFunctionWakeupPeriod	28
4.11 Parameter CanMainFunctionModePeriod	29
4.12 Parameter CanMultiplexedTransmission	29
4.13 Parameter CanTimeoutMethod	30
4.14 Parameter CanTimeoutDuration	31
4.15 Parameter CanLPduReceiveCalloutFunction	31
4.16 Parameter LPDUCalloutExtension	32

4.17 Parameter RxTimestampNotification	32
4.18 Parameter TxTimestampNotification	33
4.19 Parameter CanSetBaudrateApi	33
4.20 Parameter CanApiEnableMbAbort	34
4.21 Parameter CanAbortOnlyOneMB	34
4.22 Parameter CanLlceEnableHeadlessMode	35
4.23 Parameter CanLlceHeadlessModeUsed	35
4.24 Parameter CanCustomRxFunction	36
4.25 Parameter CanWriteCustomCallback	36
4.26 Parameter CanTxConfirmationCustomCallback	37
4.27 Parameter CanAuxFilterEnable	37
4.28 Parameter CanAuxFilterCallback	38
4.29 Parameter BusOffNotification	38
4.30 Parameter CanPublicIcomSupport	39
4.31 Reference CanEcucPartitionRef	39
4.32 Reference CanOsCounterRef	39
4.33 Reference CanSupportTTCANRef	40
4.34 Container CanMainFunctionRWPeriods	40
4.35 Parameter CanMainFunctionPeriod	42
4.36 Parameter AckInterface	42
4.37 Parameter RxInterface	43
4.38 Container CanIcomGeneral	43
4.39 Parameter CanIcomLevel	44
4.40 Parameter CanIcomVariant	44
4.41 Container CanConfigSet	45
4.42 Container CanErrorReporting	45
4.43 Parameter CAN_PROTOCOL_ERR	46
4.44 Parameter CAN_PROTOCOL_WRN	47
4.45 Parameter DATA_LOST_ERR	47
4.46 Parameter CONFIGURATION_ERR	48
4.47 Parameter INTERNAL_ERR	48
4.48 Parameter CanErrorNotif	49
4.49 Parameter CanMainFunctionErrorEn	49
4.50 Container CanController	50
4.51 Parameter CanHwChannel	50
4.52 Parameter RWDestInterface	51
4.53 Parameter CanControllerActivation	52
4.54 Parameter CanControllerBaseAddress	52
4.55 Parameter CanControllerIcomSupported	52
4.56 Parameter CanControllerId	53

4.57 Parameter CanRxProcessing	53
4.58 Parameter CanTxProcessing	54
4.59 Parameter CanBusoffProcessing	54
4.60 Parameter CanWakeupFunctionalityAPI	55
4.61 Parameter CanWakeupProcessing	55
4.62 Parameter CanWakeupSupport	56
4.63 Parameter CanTimeStampEnable	56
4.64 Parameter CanProtocolException	57
4.65 Parameter CanListenOnlyMode	57
4.66 Parameter CanLoopBackMode	58
4.67 Parameter SelfReceptionEnable	59
4.68 Parameter CanTxFifoMode	59
4.69 Parameter AutomaticBusOffRecovery	60
4.70 Parameter ManualBusOffRecovery	60
4.71 Reference CanControllerDefaultBaudrate	61
4.72 Reference CanControllerEcucPartitionRef	61
4.73 Reference CanCpuClockRef	62
4.74 Reference CanWakeupSourceRef	62
4.75 Container CanControllerBaudrateConfig	63
4.76 Parameter CanAdvancedSetting	63
4.77 Parameter CanBusLength	64
4.78 Parameter CanPropDelayTranceiver	64
4.79 Parameter CanControllerCheckCanStandard	65
4.80 Parameter CanControllerPrescaler	66
4.81 Parameter CanControllerBaudRateConfigID	67
4.82 Parameter CanControllerBaudRate	67
4.83 Parameter CanControllerSyncSeg	68
4.84 Parameter CanControllerPropSeg	68
4.85 Parameter CanControllerSeg1	69
4.86 Parameter CanControllerSeg2	69
4.87 Parameter CanControllerSyncJumpWidth	70
4.88 Container CanControllerFdBaudrateConfig	70
4.89 Parameter CanAdvancedSettingFd	70
4.90 Parameter CanBusLengthFd	71
4.91 Parameter CanControllerFDPrescaler	71
4.92 Parameter CanControllerFdBaudRate	72
4.93 Parameter CanControllerFdSyncSeg	73
4.94 Parameter CanControllerPropSeg	73
4.95 Parameter CanControllerSeg1	74
4.96 Parameter CanControllerSeg2	74

4.97 Parameter CanControllerSyncJumpWidth	74
4.98 Parameter CanControllerSspOffset	75
4.99 Parameter CanControllerTxBitRateSwitch	75
4.100 Container CanTTController	76
4.101 Parameter CanTTControllerApplWatchdogLimit	76
4.102 Parameter CanTTControllerCycleCountMax	77
4.103 Parameter CanTTControllerExpectedTxTrigger	77
4.104 Parameter CanTTControllerExternalClockSynchronisation	78
4.105 Parameter CanTTControllerGlobalTimeFiltering	78
4.106 Parameter CanTTControllerInitialRefOffset	79
4.107 Parameter CanTTControllerInterruptEnable	79
4.108 Parameter CanTTControllerLevel2	79
4.109 Parameter CanTTControllerNTUConfig	80
4.110 Parameter CanTTControllerOperationMode	80
4.111 Parameter CanTTControllerSyncDeviation	81
4.112 Parameter CanTTControllerTURRestore	81
4.113 Parameter CanTTControllerTimeMaster	82
4.114 Parameter CanTTControllerTimeMasterPriority	82
4.115 Parameter CanTTControllerTxEnableWindowLength	83
4.116 Parameter CanTTControllerWatchTriggerGapTimeMark	83
4.117 Parameter CanTTControllerWatchTriggerTimeMark	84
4.118 Parameter CanTTIRQProcessing	84
4.119 Reference CanTTControllerEcucPartitionRef	85
4.120 Container CanHardwareObject	85
4.121 Parameter Dummy_HRH	86
4.122 Parameter CanFdPaddingValue	86
4.123 Parameter CanHandleType	86
4.124 Parameter CanIdType	87
4.125 Parameter CanObjectId	87
4.126 Parameter CanObjectType	88
4.127 Parameter CanTxAddFrameMac	89
4.128 Parameter CanLlceKeyHandle	89
4.129 Parameter CanHardwareObjectUsesPolling	90
4.130 Parameter CanTriggerTransmitEnable	90
4.131 Parameter CanHwObjectCount	90
4.132 Parameter CanObjectPayloadLength	91
4.133 Reference CanControllerRef	92
4.134 Reference CanMainFunctionRWPeriodRef	92
4.135 Container CanHwFilter	92
4.136 Parameter CanHwFilterCode	94

4.137 Parameter CanHwFilterMask	94
4.138 Container RangeFilter	95
4.139 Parameter RangeStart	96
4.140 Parameter RangeEnd	96
4.141 Container CanAdvancedFeature	97
4.142 Reference CanAdvancedFeatureRef	97
4.143 Container CanTTHardwareObjectTrigger	98
4.144 Parameter CanTTHardwareObjectBaseCycle	98
4.145 Parameter CanTTHardwareObjectCycleRepetition	98
4.146 Parameter CanTTHardwareObjectTimeMark	99
4.147 Parameter CanTTHardwareObjectTriggerId	99
4.148 Parameter CanTTHardwareObjectTriggerType	100
4.149 Container CanAuxFilter	100
4.150 Parameter CanIdType	101
4.151 Parameter Filter_Type	101
4.152 Parameter AuxSearchResult	102
4.153 Parameter CanHwFilterCode	102
4.154 Parameter CanHwFilterMask	103
4.155 Parameter RangeStart	104
4.156 Parameter RangeEnd	104
4.157 Reference CanControllerRef	105
4.158 Container CanIcom	105
4.159 Container CanIcomConfig	105
4.160 Parameter CanIcomConfigId	106
4.161 Parameter CanIcomWakeOnBusOff	106
4.162 Container CanIcomWakeupCauses	107
4.163 Container CanIcomRxMessage	107
4.164 Parameter CanIcomCounterValue	108
4.165 Parameter CanIcomMessageId	108
4.166 Parameter CanIcomIdOperation	109
4.167 Parameter CanIcomMessageIdMask	110
4.168 Parameter CanIcomMissingMessageTimerValue	110
4.169 Parameter CanIcomPayloadLengthError	111
4.170 Container CanIcomRxMessageSignalConfig	111
4.171 Parameter CanIcomSignalMask	112
4.172 Parameter CanIcomSignalOperation	112
4.173 Parameter CanIcomSignalValue	113
4.174 Parameter DLCLowValue	113
4.175 Parameter DLCHighValue	114
4.176 Reference CanIcomSignalRef	114

4.177 Container CommonPublishedInformation	115
4.178 Parameter ArReleaseMajorVersion	115
4.179 Parameter ArReleaseMinorVersion	116
4.180 Parameter ArReleaseRevisionVersion	116
4.181 Parameter ModuleId	117
4.182 Parameter SwMajorVersion	117
4.183 Parameter SwMinorVersion	118
4.184 Parameter SwPatchVersion	118
4.185 Parameter VendorApiInfix	119
4.186 Parameter VendorId	120



Chapter 1

Revision History

Revision	Date	Author	Description
1.0	29.09.2023	NXP LLCE Team	Prepared for release S32G LLCE AUTOSAR 4.4 Version 1.0.7

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductor AUTOSAR Controller Area Network on LLCE (CAN_43_LLCE) for S32.

AUTOSAR CAN_43_LLCE driver configuration parameters and deviations from the specification are described in CAN_43_LLCE Driver chapter of this document. AUTOSAR CAN_43_LLCE driver requirements and APIs are described in the AUTOSAR CAN_43_LLCE driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
DEM	Diagnostic Event Manager
DET	Development Error Tracer
C/CPP	C and C++ Source Code
VLE	Variable Length Encoding
N/A	Not Applicable
MCU	Micro Controller Unit

2.5 Reference List

#	Title	Version
1	Specification of CAN_43_LLCE Driver	AUTOSAR Release 4.4.0
2	S32G2 Reference Manual	Rev. 5, 05/2022
3	S32G2 Data Sheet	Rev. 5, 05/2022
4	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 0P77B, Rev. 3.1
5	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 1P77B, Rev. 1.1
6	S32G3 Reference Manual	Rev. 2, 09/2022
7	S32G3 Data Sheet	Rev. 2, Draft C, 09/2021
8	S32G3 Errata Document	S32G3 Mask Set Errata for Mask 1P72B, Rev. 2.2

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware resources](#)
- [Deviation from Requirements](#)
- [Driver Limitations](#)
- [Driver Usage and Configuration Tips](#)
- [Runtime Errors](#)
- [Symbolic Names Disclaimer](#)

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.4 Rev0000 CAN Driver Software Specification document (See Table [Reference List](#)).

3.2 Driver Design Summary

The LLCE module contains 16 BCAN blocks, each individually configurable. All BCANs are controlled by the LLCE unit, which provides state management and buffering.

BCAN features:

Full implementation of the CAN with Flexible Data Rate (CAN FD) and Classical CAN (CAN 2.0 B)
Standard ID frames
Extended ID frames
Zero to sixty-four bytes of data length
Programmable bit rate
Compliant with ISO 11898-1 standard
Listen-Only mode capability
Independence from the transmission medium
Transceiver Delay Compensation when transmitting data phase of CAN FD frames
CRC status for transmitted and received messages

LLCE aims to provide:

Low latency processing of communication interface
Offloading of host CPU for all interface related tasks
Direct data transfer to/ from HSE for security related tasks

3.3 Hardware resources

The platform supports 16 BCAN controllers.

3.4 Deviation from Requirements

The driver deviates from the AUTOSAR CAN_43_LLCE Driver software specification in some places.

There are also some additional requirements (on top of requirements detailed in AUTOSAR CAN_43_LLCE Driver software specification) which need to be satisfied for correct operation.

1. LPduReceiveCalloutFunction prototype can be optionally extended to allow reporting of HRH number greater than 127, thus changing the API prototype stated by SWS_Can_00443 from "uint8 Hrh" to "uint16 Hrh".
2. The driver depends of the OS to get a reference value for timeout calculation.
3. For the blocking functions, the timeout duration is given in seconds.
4. CanOsCounterRef is not used anymore, the using of OsCounter is done by selecting CanTimeoutMethod to OSIF_COUNTER_SYSTEM.
5. The base address for the controllers is not user input.
6. The Can driver supports only 6 APIs of Can_MainFunction_Write() for processing transmitted MBs.
The APIs obey the following pattern:
Can_MainFunction_Write_0(), Can_MainFunction_Write_1(), ... and until Can_MainFunction_Write_↵
10(). .
7. The Can driver supports only 6 APIs of Can_MainFunction_Read() for processing received MBs.
The APIs obey the following pattern:
Can_MainFunction_Read_0(), Can_MainFunction_Read_1(), ... and until Can_MainFunction_Read_10().
8. In case of bus-off processing, the CAN_LLCE driver can be configured optionally to
 - auto-recovery, which allows the firmware to resync the controller on the bus when bus-off occurs.
 - manual-recovery, which activates the Can_43_LLCE_ManualBusOffRecovery() API (Non Autosar) and allows the user to recover the controller from bus-off discretionary. No outstanding TX or RX frames are discarded.

Table provides Status column description.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out

of scope for the CAN_43_LLCE driver.

Requirement	Status	Description	Notes
SWS_Can_00242	N/S	Icu_DisableNotification shall be called when "external" Can controllers have been transitioned to STOPPED state (CAN_CS_STOPPED).	All controllers are on chip.
SWS_Can_00242	N/S	If an off-chip CAN controller is used, the driver uses services of other MCAL drivers (i.e. SPI). These drivers need to be up and running before the CAN controller can be initialized. The sequence of initialization of different drivers is partly specified in [7]. Only Synchronous APIs may be used because the CAN driver does not provide callback functions that can be called by the MCAL driver. Thus the type of connection between μ C and CAN Hardware Unit has only impact on implementation and not on the API.	No off-chip controller is used.
SWS_Can_00110	N/S	There is no requirement regarding the execution order of the CAN main processing functions.	Application Code Requirement.
SWS_Can_00240	N/S	The Can module's environment shall make sure that the Mcu module is initialized before initializing the Can module.	Can driver cannot access any variable of Mcu module for checking the state.
SRS_BSW_00389_Conf	N/S	Specifies the CAN controller base address.	The possibility for user to input the base address of the controller is not supported by the Can Driver. The base address is given in the Base module.
SWS_Can_00186	N/S	If default error detection for the Can module is enabled: The function Can_MainFunction_Wakeup shall raise the error CAN_E_UNINIT if the driver is not yet initialized.	The external application shall call Can_MainFunction_Wakeup only after driver initialization. Only implemented in platform which support wake up feature.
SWS_Can_00242	N/S	If an off-chip CAN controller is used, the Can module shall use services of other MCAL drivers (e.g. SPI).	All current controllers are onchip.
SWS_Can_00244	N/S	The Can module shall use the synchronous APIs of the underlying MCAL drivers and shall not provide callback functions that can be called by the MCAL drivers.	All current controllers are onchip. No callback can be called by other drivers.

Requirement	Status	Description	Notes
SWS_Can_00391	N/S	Can module implementations for off-chip CAN controllers shall include the header file Spi.h. By this inclusion, the APIs to access an external CAN controller by the SPI module [12] are included.	Controllers are onchip.
SWS_Can_00257	N/S	When the CAN hardware supports sleep mode and is triggered to transition into SLEEP state, the Can module shall set the controller to the SLEEP state from which the hardware can be woken over CAN Bus.	Controller not support sleep mode.
SWS_Can_00265	N/S	The function Can_SetControllerMode(CAN_T_SLEEP) shall set the controller into sleep mode.	Only applicable for platform support hardware wake-up. Otherwise, only logical sleep is implemented.
SWS_Can_00270	N/S	On hardware wakeup (triggered by a wake-up event from CAN bus), the CAN controller shall transition into the state STOPPED.	Platform does not support a WAKE UP mode.
SWS_Can_00271	N/S	On hardware wakeup (triggered by a wake-up event from CAN bus), the Can module shall call the function EcuM_CheckWakeup either in interrupt context or in the context of Can_MainFunction_Wakeup.	Platform does not support a WAKE UP mode.
SWS_Can_00269	N/S	The Can module shall not further process the L-PDU that caused a wake-up.	Platform does not support a WAKE UP mode.
SWS_Can_00048	N/S	In case of a CAN bus wake-up during sleep transition, the function Can_SetControllerMode(CAN_T_WAKEUP) shall return CAN_NOT_OK.	For HW not support on-chip wakeup, this event will not occur.
SWS_Can_00364	N/S	If the ISR for wakeup events is called, it shall call EcuM_CheckWakeup in turn. The parameter passed to EcuM_CheckWakeup shall be the ID of the wakeup source referenced by the CanWakeupSourceRef configuration parameter.	Only for platform support wake up.
SWS_Can_00461	N/S	If hardware supports wake-up (i.e. CanWakeupSupport == true), it shall be checked during controller initialization if there was a wake-up event on the specific CAN controller. If a wake-up event has been detected, the wake-up shall directly be reported to the EcuM via EcuM_SetWakeupEvent callback function.	Platform does not support WAKE UP mode.

Requirement	Status	Description	Notes
SWS_Can_00294	N/S	The function Can_SetController↔ Mode shall disable the wake-up interrupt, while checking the wake-up status.	Platform does not support a WAKE UP mode.
SWS_Can_00361	N/S	The function Can_CheckWakeup shall check if the requested CAN controller has detected a wakeup. If a wakeup event was successfully detected, reporting shall be done to EcuM via API EcuM_SetWakeup↔ Event.	AAI-279: reject this requirement because our SoC does not support wakeup.
SWS_Can_00445	N/S	Can driver shall use the following APIs provided by Icu driver, to enable and disable the wakeup event notification:Icu_↔ EnableNotificationIcu_Disable↔ Notification.	Not implemented for OnChip platform.
SWS_Can_00446	N/S	Icu_EnableNotification shall be called when "external" Can controllers have been transitioned to SLEEP state (CAN_CS_SLEEP).	Not implemented for OnChip platform.
SWS_Can_00110	N/S	There is no requirement regarding the execution order of the CAN main processing functions.	this is not a requirement.
SWS_Can_00228	N/S	The Can_MainFunction_Wakeup function performs the polling of wake-up events that are configured statically as 'to be polled'.	Not implemented for platform not support hardware wake-up.
SWS_Can_00112	N/S	The function Can_MainFunction↔ _Wakeup shall perform the polling of wake-up events that are configured statically as 'to be polled'.	Not implemented for platform not support hardware wake-up.
SWS_Can_00185	N/S	The Can module may implement the function Can_MainFunction↔ _Wakeup as empty define in case no polling at all is used.	AAI-279: reject this requirement because our SoC does not support wakeup.
SWS_Can_00999	N/S	These requirements are not applicable to this specification.	this is not a requirement.
ECUC_Can_00357	N/S	Name CanMainFunctionWakeup↔ Period Description This parameter describes the period for cyclic call to Can_MainFunction_Wakeup. Unit is seconds.	Shall be supported only by platforms which include hardware wake-up supported.
ECUC_Can_00430	N/S	Name CanSupportTTCANRef Description The parameter refers to CanIfSupportTTCAN parameter in the CAN Interface Module configuration. The CanIfSupportTTCAN parameter defines whether TTCAN is supported.	This is a HW limitation, some platform does not support TTCAN feature.

Requirement	Status	Description	Notes
ECUC_Can_00382	N/S	Name CanControllerBaseAddress Description Specifies the CAN controller base address.	Controller base address cannot be configured.
ECUC_Can_00466	N/S	Name CanWakeupFunctionalityA↔ PI Description Adds / removes the service Can_CheckWakeup() from the code. True: Can_Check↔Wakeup can be used. False: Can↔_CheckWakeup cannot be used.	Shall be supported only by platforms which include hardware wake-up supported
ECUC_Can_00330	N/S	Name CanWakeupSupport Descrip- tion CAN driver support for wakeup over CAN Bus.	Shall be supported only by platforms which include hardware wake-up supported. Always has value = false
ECUC_Can_00359	N/S	CanWakeupSourceRef Description This parameter contains a reference to the Wakeup Source for this controller as defined in the E↔CU State Manager. Implementation Type: reference to EcuM↔_WakeupSourceType Multiplicity 0..1 Type Symbolic name reference to [EcuMWakeupSource] Post-↔Build Variant Multiplicity false Post-Build Variant Value false Multiplicity Configuration Class Pre-compile time X All Variants Link time – Post-build time – Value Configuration Class Pre-compile time X All Variants Link time – Post-build time – Scope / Dependency scope: local Included Containers Container Name Multiplicity Scope / Dependency CanControllerBaudrate↔Config 1..* This container contains bit timing related configuration parameters of the CAN controller(s). CanTTController 0..1 CanTTController is specified in the SWS TTCAN and contains the configuration parameters of the TTC↔AN controller(s) (which are needed in addition to the configuration parameters of the CAN controller(s)). This container is only included and valid if TTCAN is supported by the controller, enabled (see Can↔SupportTTCANRef, ECUC_Can↔_00430), and used.	This requirement is only applicable for platform support HW wake up.

Requirement	Status	Description	Notes
ECUC_Can_00440	N/S	Container Name CanIcom Description This container contains the parameters for configuring pretended networking Configuration Parameters Included Containers Container Name Multiplicity Scope / Dependency CanIcomConfig 1..* This container contains the configuration parameters of the ICOM Configuration. 10.2.12	This platform does not support Pretended Networking feature.
ECUC_Can_00459	N/S	Container Name CanIcomConfig Description This container contains the configuration parameters of the ICOM Configuration.	This platform does not support Pretended Networking feature.
ECUC_Can_00441	N/S	Name CanIcomConfigId Description: This parameter identifies the ID of the ICOM configuration.	This platform does not support Pretended Networking feature.
ECUC_Can_00442	N/S	Name CanIcomWakeOnBusOff Description: This parameter defines that the MCU shall wake if the bus off is detected or not.	This platform does not support Pretended Networking feature.
ECUC_Can_00444	N/S	Container Name CanIcomGeneral Description This container contains the general configuration parameters of the ICOM Configuration.	This platform does not support Pretended Networking feature.
ECUC_Can_00445	N/S	Name CanIcomLevel Description Defines the level of Pretended Networking. This parameter is reserved for future implementations (Pretended Networking level 2).	This platform does not support Pretended Networking feature.
ECUC_Can_00446	N/S	Name CanIcomVariant Description Defines the variant, which is supported by this CanController	This platform does not support Pretended Networking feature.
ECUC_Can_00447	N/S	Container Name CanIcomRx↔ Message Description This container contains the configuration parameters for the wakeup causes for matching received messages. It has to be configured as often as received messages are defined as wakeup cause. constraint: For all CanIcomRxMessage instances the Message IDs which are defined in CanIcomMessageId and in Can↔IcomRxMessageIdMask shall not overlap.	This platform does not support Pretended Networking feature.

Requirement	Status	Description	Notes
ECUC_Can_00448	N/S	Name CanIcomCounterValue Description This parameter defines that the MCU shall wake if the message with the ID is received n times on the communication channel.	This platform does not support Pretended Networking feature.
ECUC_Can_00449	N/S	Name CanIcomMessageId Description This parameter defines the message ID the wakeup causes of this CanIcomRxMessage are configured for. In addition a mask (CanIcomMessageIdMask) can be defined, in that case it is possible to define a range of rx messages, which can create a wakeup condition.	This platform does not support Pretended Networking feature.
ECUC_Can_00465	N/S	Name CanIcomMessageIdMask Description Describes a mask for filtering of CAN identifiers. The CAN identifiers of incoming messages are masked with this CanIcomMessageIdMask. If the masked identifier matches the masked value of CanIcomMessageId, it can create a wakeup condition for this CanIcomRxMessage. Bits holding a 0 mean don't care, i.e. do not compare the message's identifier in the respective bit position. The mask shall be build by filling with leading 0.	This platform does not support Pretended Networking feature.
ECUC_Can_00450	N/S	Name CanIcomMissingMessageTimerValue Description This parameter defines that the MCU shall wake if the message with the ID is not received for a specific time in s on the communication channel.	This platform does not support Pretended Networking feature.
ECUC_Can_00451	N/S	Name CanIcomPayloadLengthError Description This parameter defines that the MCU shall wake if a payload error occurs	This platform does not support Pretended Networking feature.

Requirement	Status	Description	Notes
ECUC_Can_00452	N/S	Container Name CanIcomRx↔ MessageSignalConfig Description This container contains the configuration parameters for the wakeup causes for matching signals. It has to be configured as often as a signal is defined as wakeup cause. If at least one Signal conditions defined in a CanIcomRxMessageSignal↔ Config evaluates to true or if no CanIcomRxMessageSignalConfig are defined, the whole wakeup condition is considered to be true. All instances of this container refer to the same frame/pdu (see CanIcomMessageId).	This platform does not support Pretended Networking feature.
ECUC_Can_00487	N/S	Name CanIcomSignalMask Description This parameter shall be used to mask a signal in the payload of a CAN message. The mask is binary AND with the signal payload. The result will be used in combination of the operations defined in CanIcomSignalOperation with the CanIcomSignalValue.	This platform does not support Pretended Networking feature.
ECUC_Can_00462	N/S	Name CanIcomSignalOperation Description This parameter defines the operation, which shall be used to verify the signal value creates a wakeup condition.	This platform does not support Pretended Networking feature.
ECUC_Can_00488	N/S	Name CanIcomSignalValue Description This parameter shall be used to define a signal value which shall be compared (CanIcom↔ SignalOperation) with the masked CanIcomSignalMask value of the received signal (CanIcomSignal↔ Ref).	This platform does not support Pretended Networking feature.
ECUC_Can_00456	N/S	Name CanIcomSignalRef Description This parameter defines a reference to the signal which shall be checked additional to the message id (CanIcomMessageId). This reference is used for documentation to define which ComSignal originates this filter setting. All signals being referred by this reference shall point to the same PDU.	This platform does not support Pretended Networking feature.

Requirement	Status	Description	Notes
ECUC_Can_00443	N/S	Container Name CanIcom↵ WakeupCauses Description This container contains the configuration parameters of the wakeup causes to leave the power saving mode. Configuration Parameters	This platform does not support Pretended Networking feature.
CPR_RTD_00420	N/S	If DET error reporting is enabled, the CAN_LLCE will check upon each API call if the requested resource is configured to be available on the current core, and in case of error will return CAN_LLCE_E↵ _PARAM_CONFIG.	Not in scope of the current release.

Can_43_LLCE_PBcfg_VS.c files will contain the definition for all parameters that are variant aware, independent of the configuration class that will be selected (PC, LT, PB).

Can_43_LLCE_Cfg.c file will contain the definition for all parameters that are not variant aware.

3.4.1 LLCE Headless Mode

The Can_43_LLCE driver supports usage of the so-called LLCE Headless Mode, allowing LLCE to boot and start working before the AUTOSAR application is fully active. For more details regarding LLCE Headless Mode, see the LLCE Firmware Guide.

To enable headless mode support in the driver, activate the relevant options in the configurator. There is a global option that will affect code execution, and a per-variant option that allows the user to use headless mode or not depending on variant. For variants which use headless mode, the generator will create an additional pair of files named Can_43_LLCE_<variant>_PBcfg.h/c for each variant. These files are to be used to create the headless configuration, and shall not be compiled in the user application.

When headless mode is enabled, certain steps during init are skipped. This avoids generating errors by not re-initializing the LLCE firmware and the active channels. This may cause unexpected behavior in some cases. For example, the first START command for a controller is skipped in Headless Mode, but STOP commands are not. So issuing STOP commands before START would cause the controllers to be stopped, and then not started again on the first attempt.

3.5 Driver Limitations

- The Can driver supports only 6 API instances of Can_MainFunction_Write() for processing transmitted M↵Bs. The APIs name obey the following pattern: Can_MainFunction_Write_'CanMainFunctionRWPeriods.ShortName'().
- The Can driver supports only 6 API instances of Can_MainFunction_Read() for processing received MBs. The APIs name obey the following pattern: Can_MainFunction_Read_'CanMainFunctionRWPeriods.Short↵Name'().
- The Can driver supports up to 4096 reception message buffers, including 1700 buffers of 64 bytes size and 2396 buffers of 8 bytes size.
- The Can driver supports up to 512 Tx acknowledgment buffers which can be allocated in any way across all the 16 channels.

- Flexible allocation of the reception buffers and the acknowledgement buffers along all the 16 channels
- The Can driver supports the configuration of a maximum of 1024 filters on S32G2, of which up to 512 can be BASIC filters (configurable); and 4096 filters on S32G3 of any type.
- The Can driver can be deployed on maximum 2 cores simultaneously with full access to the exposed services, limited by the LLCE firmware. In case of multicore usage, the allocation of controllers to host cores is flexible, as long as the restriction of FIFO interface (RX/ACK interface field in CanController container) is correctly followed with the correct ranges, as follows: Host0 must use the FIFO interfaces 0-7 (any controller can subscribe to any of this interfaces), Host1 must use the FIFO interfaces range 8-15. This restricted allocation comes from the interrupt routing of the FIFOs used as interface between LLCE firmware and CAN driver, which shares the same interrupt line along 8 FIFOs. Each host interface works with groups of 8 FIFOs to subscribe to this rule. The default settings presume that each controller is assigned to its unique and corresponding FIFO interface with the same number.

3.6 Driver Usage and Configuration Tips

This chapter describes how to configure for advanced features which are not described by AUTOSAR SWS (i.e NON-ASR features).

- Busoff is implemented according to ISO_11898. In Autosar context, LLCE notifies (in interrupt/polling as per configuration) the CAN driver about busoff condition of a specific controller. The CAN driver discards the eventual pending RX messages, reports the confirmations received before busoff, and notifies in turn CanIf using the u8CanIfCtrlId. The CanIf reaction for getting the interface back to active is to call SetController↔ Mode (start) on that interface. In Non-Autosar context, there is an additional option (in configurator, or as host interface option) as reaction to busoff event, namely auto-recovery. LLCE firmware is able to recover automatically from bus-off state after 128 occurrences of 11 recessive bits at nominal bitrate, when a special notification LLCE_NOTIF_BUSOFF_DONE is issued to the host. The pending transmissions are dropped as reaction to busoff detection. In case a stop request is issued between bus-off and bus-off done, the bus-off done will not be notified to host up until a new start command is requested.

3.6.1 Driver APIs usage

- Can_Llce_GetFwVersion is an API which reports the firmware version in the form of a string embedding a symbol for each functionality described in detail in firmware user guide.
- GetFwComponentVersion is an API which reports the firmware version in the form of a string embedding a symbol for each functionality described in detail in firmware user guide.
- Can_Llce_GetControllerStatus is an API which reports the current status of a specific controller, by querying LLCE firmware.
- Can_Llce_GetControllerErrorState is an API which reports the possible errors occurred in BCAN controller during its operation, by querying LLCE firmware.
- Can_Llce_ProcessNotificationISR is an API which reports in interrupt any notifications raised by LLCE firmware about runtime events, like bus-off or datalost.
- Can_Llce_GetFwComponentVersion is an API which reports the version of a specific firmware component.
- Can_43_LLCE_Write is the only thread-safe service. ARM CM7 is able to do unaligned memory accesses, thus SduDataPtr which comes from AUTOSAR stack as a pointer of the uint8* type is not a problem of unaligned accesses.

- An address can be freed either by calling the `RemoveFilter` function or leaving extra free filters via host configuration.
- `Can_43_LLCE_AbortMb()` API (Non Autosar) is defined if this feature is enabled by `CanApiEnableMbAbort`.
- `Can_LLCE` driver supports loop back mode to verify driver internally. In this case, it is necessary to configure a filter in order to receive the transmit message.
- `Can_43_LLCE_ManualBusOffRecovery()` API (Non Autosar) - this synchronous service requests bus-off recovery when the controller is configured in manual bus-off recovery mode. No outstanding TX or RX frames are discarded.

The below callbacks help the user identify a TX MB in memory when the ACK for that frame is received from LLCE. The risk is when calling `Can_Write` in a MB that was freed by LLCE, along with its index, but the ACK was not read in time, thus the callback is invoked later with the index of the MB already overwritten by `Can_Write` service.

- `Can_43_LLCE_CustomWriteCallback` is called between when `Can_43_LLCE_Write` is called, and the data is pushed to LLCE.
- `Can_43_LLCE_CustomTxConfirmationCallback` is called when a `TxAck` is received from LLCE and processed. Filter setup at runtime (after `Can_43_LLCE_Init()` phase):
- In order to set new filters at runtime (after `Can_43_LLCE_Init()` phase), the user shall first remove a filter by calling `Can_43_LLCE_RemoveFilter()`, thus creating a free address for new filter. This is due to the statical allocation of the resources along the controllers with the exact number of filters generated.
- Next step is to call either `Can_43_LLCE_SetFilter()`, which set the new filter at the first free location found; or `Can_43_LLCE_SetFilterAtAddress()` which set the new filter at a given address (typically the address returned by `Can_43_LLCE_RemoveFilter()`). From optimization perspective, it is recommended to use the `Can_43_LLCE_SetFilterAtAddress()` so that LLCE firmware will not search for the first empty filter location.
- Filters order in configuration : S32G2 offers: First 512 filters are exact match filters, next 512 are configurable. Overlaying filters shall ideally not exist, otherwise the matching filter might not be the one expected. The matching filter is always the one with the lowest address, thus an exact match will match before a configurable (mask/range) one, no matter the order of filters defined in configuration. In case of multiple overlaying configurable filters, the first one that matches is the one that was first passed to LLCE firmware.
- For S32G3 platform, all 4096 available filters are configurable, and the same rule for matching order applies.
- Range filter type : Specify that this filter is of range type. This over-rides the information in the standard `CanHwFilter`. Being a non Autosar concept, the range filter discards the information of `CanHandleType` and `CanIdType`. If enabled, the filter will accept IDs from `RangeStart` to `RangeEnd` inclusively. Unlike masked filters, the range filter does not work in MIXED mode. The user shall specify whether the range start, end or both are extended IDs by manually setting the IDE bit (apply `0x40000000U` mask over the desired ID). The range filter type allows Standard Can ID type for Range Start and Extended Can ID type for Range End.

3.6.2 Can Hardware Object Handle (HOH) configuration

The `CanHardwareObject` container in Tresos plugin is used to configure the operating for Rx and Tx MBs.

Since the Can Controllers can be configured in MIXED mode, multiple HTHs and/or HRHs of the same controller can be configured in any mode: interrupt or polling.

For HRHs, configuring a reference to an advanced feature allows configuring new rules for advanced features in `LLCE_AF` plugin, like routing CAN to CAN with different frame conversion types, CAN to ETH, logging (mirroring) to a different core, frame authentication.

3.6.3 Configuration Tips

- Any filter uses an entry in RX-LUT (accelerator used for accepting a CAN frame based on its frame ID and Hardware Object configuration). The filters marked as FULL are 'exact match' (as AUTOSAR describes: only one identifier is handled by the filter) and the filter mask field ('Can Hw Filter Mask') is omitted because it is implicitly 0xFFFFFFFF, whereas the filters marked as BASIC are masked (as AUTOSAR describes: several identifiers are handled by the same filter) and they use the 'configurable entries' in RX-LUT. The final purpose of the mask filters is to comprise more exact match filters and to efficiently use the resources.
- Each Hardware object used for RECEIVE specifies an interface for receiving new CAN frames. If the channels are set in interrupt mode, the interface number corresponds to the channel number; if the channel is set in polling, the host will receive new CAN frames through a special fifo, corresponding to the polling class which has been set for that hardware object in the CanMainFunctionRWPeriods list.
- The default FIFO interface number for each controller is the hw controller number (E.g FIFO_INTERFACE_3 for BCAN_3 peripheral). A custom configuration can set any of the FIFO interface to any of the channel in the CanController container. One reason to do this is for the flexible allocation of any physical CAN controller to any of the 2 host cores by using the FIFO range 0-7 for one host and FIFO range 8-15 for the other host, due to the interrupt line routing in groups of 8 FIFOs. The FIFO_INTERFACE field is a mapping layer which allows the flexible assignment of controllers to a core, in context of multicore usage, for up to 2 cores. This features allows grouping of any CAN controllers to the same core based on common FIFO interface selection. If multicore is not used, or interrupt support is not used at all in a multicore environment for host interface, it is not necessary to overwrite the RX/ACK interface field.
- A controller can have 1 or more hardware objects. If interrupt chosen, then all the frames accepted by its hardware objects will be reported through interrupt (first 16 interfaces). If polling, all its filters are linked to a polling class and all frames accepted by these filters will be sent to polling interfaces and the user application will periodically call Main_Function_Read(polling_class) to get the frames corresponding to that polling class. If mixed mode selected for the controller, its filters can be in either interrupt or polling, meaning that the frames accepted by a hardware object will be reported through interrupt, and the frames accepted by another hardware object which is linked to a polling class (RWPeriodReference) will be reported through polling and the user application can get them anytime by calling MainFunction_Read. It's possible that different controllers to have hardware objects which share the same polling class and a call to MainFunction_Read will return all the new received frames.
- All filter entries for a channel are searched in 'order' – the first match completes the search. Therefore, If FULL filter is tried first and it fails, search will continue and eventually match a BASIC filter.
- Range filter type : Specify that this filter is of range type. This over-rides the information in the standard Can↔HwFilter.
- Baudrate configuration : Controller Prescaler and Controller Prescaler FD should have the same value to minimize the errors. The sample point should be around 75-85% ideally.
- CanControllerActivation : Defines if a CAN controller is used in the configuration. When this is disabled, the controller will not be initialized and cannot be started. Moreover, the HOHs configuration corresponding to the disabled controller will not be passed to LLCE firmware.

3.6.4 LLCE Headless Mode

See section under "Deviation from requirements" for details on the use of Headless Mode using this driver.

3.7 Runtime Errors

The driver generates the following DET errors at runtime.

Function	Error Code	Condition triggering the error
All functions	CAN_43_LLCE_E_UNINIT	The driver is not yet initialized
Can_43_LLCE_Write	CAN_43_LLCE_E_PARAM_↔ DATA_LENGTH	The length is more than 8 bytes (non-FD) or 64 bytes (FD)
Can_43_LLCE_CheckWakeup Can_43_LLCE_SetBaudrate Can_43_LLCE_GetController↔ ErrorState Can_43_LLCE_↔ SetControllerMode Can_43_L↔ LCE_GetControllerMode Can↔ _43_LLCE_DisableController↔ Interrupts Can_43_LLCE_↔ EnableControllerInterrupts Can↔ _43_LLCE_GetControllerRx↔ ErrorCounter Can_43_LLCE_↔ GetControllerTxErrorCounter	CAN_43_LLCE_E_PARAM_↔ CONTROLLER	The parameter Controller is out of range
Can_43_LLCE_Init	CAN_43_LLCE_E_PARAM_↔ POINTER	The parameter Config is a null-pointer
Can_43_LLCE_GetController↔ Mode	CAN_43_LLCE_E_PARAM_↔ POINTER	The parameter ControllerModePtr is a null-pointer
Can_43_LLCE_Write	CAN_43_LLCE_E_PARAM_↔ POINTER	The parameter PduInfo or the SDU pointer inside PduInfo (with Trigger Transmit disabled) is a null-pointer
Can_43_LLCE_GetVersionInfo	CAN_43_LLCE_E_PARAM_↔ POINTER	The paramete versionInfo is a null pointer
Can_43_LLCE_Init, Can_43_L↔ LCE_SetBaudRate	CAN_43_LLCE_E_TRANSITI↔ ON	The controller is not in CAN_CS↔ _STOPPED
Can_Llce_SetcontrollerToSleep↔ Mode, Can_Llce_Setcontroller↔ ToStartMode	CAN_43_LLCE_E_TRANSITI↔ ON	The controller is not in STOP state
Can_Llce_SetcontrollerMode	CAN_43_LLCE_E_TRANSITI↔ ON	A invalid transition has been requested
Can_Llce_SetcontrollerToStop↔ Mode	CAN_43_LLCE_E_TRANSITI↔ ON	The controller is not in START or SLEEP state
Can_Llce_ProcessNotificationISR	CAN_43_LLCE_E_DATALOST	The firmware reports detections of an overrun or overwrite event

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Can](#)
 - Container [CanGeneral](#)
 - * Parameter [CanDevErrorDetect](#)
 - * Parameter [CanEnableUserModeSupport](#)
 - * Parameter [CanMulticoreSupport](#)
 - * Parameter [HostInterface](#)
 - * Parameter [CanVersionInfoApi](#)
 - * Parameter [CanIndex](#)
 - * Parameter [CanMainFunctionBusoffPeriod](#)
 - * Parameter [CanMainFunctionWakeupPeriod](#)
 - * Parameter [CanMainFunctionModePeriod](#)
 - * Parameter [CanMultiplexedTransmission](#)
 - * Parameter [CanTimeoutMethod](#)
 - * Parameter [CanTimeoutDuration](#)
 - * Parameter [CanLPduReceiveCalloutFunction](#)
 - * Parameter [LPDUCalloutExtension](#)
 - * Parameter [RxTimestampNotification](#)
 - * Parameter [TxTimestampNotification](#)
 - * Parameter [CanSetBaudrateApi](#)
 - * Parameter [CanApiEnableMbAbort](#)
 - * Parameter [CanAbortOnlyOneMB](#)
 - * Parameter [CanLlceEnableHeadlessMode](#)
 - * Parameter [CanLlceHeadlessModeUsed](#)
 - * Parameter [CanCustomRxFunction](#)
 - * Parameter [CanWriteCustomCallback](#)
 - * Parameter [CanTxConfirmationCustomCallback](#)
 - * Parameter [CanAuxFilterEnable](#)
 - * Parameter [CanAuxFilterCallback](#)
 - * Parameter [BusOffNotification](#)
 - * Parameter [CanPublicIcomSupport](#)

- * Reference [CanEcucPartitionRef](#)
- * Reference [CanOsCounterRef](#)
- * Reference [CanSupportTTCANRef](#)
- * Container [CanMainFunctionRWPeriods](#)
 - Parameter [CanMainFunctionPeriod](#)
 - Parameter [AckInterface](#)
 - Parameter [RxInterface](#)
- * Container [CanIcomGeneral](#)
 - Parameter [CanIcomLevel](#)
 - Parameter [CanIcomVariant](#)
- Container [CanConfigSet](#)
 - * Container [CanErrorReporting](#)
 - Parameter [CAN_PROTOCOL_ERR](#)
 - Parameter [CAN_PROTOCOL_WRN](#)
 - Parameter [DATA_LOST_ERR](#)
 - Parameter [CONFIGURATION_ERR](#)
 - Parameter [INTERNAL_ERR](#)
 - Parameter [CanErrorNotif](#)
 - Parameter [CanMainFunctionErrorEn](#)
 - * Container [CanController](#)
 - Parameter [CanHwChannel](#)
 - Parameter [RWDestInterface](#)
 - Parameter [CanControllerActivation](#)
 - Parameter [CanControllerBaseAddress](#)
 - Parameter [CanControllerIcomSupported](#)
 - Parameter [CanControllerId](#)
 - Parameter [CanRxProcessing](#)
 - Parameter [CanTxProcessing](#)
 - Parameter [CanBusoffProcessing](#)
 - Parameter [CanWakeupFunctionalityAPI](#)
 - Parameter [CanWakeupProcessing](#)
 - Parameter [CanWakeupSupport](#)
 - Parameter [CanTimeStampEnable](#)
 - Parameter [CanProtocolException](#)
 - Parameter [CanListenOnlyMode](#)
 - Parameter [CanLoopBackMode](#)
 - Parameter [SelfReceptionEnable](#)
 - Parameter [CanTxFifoMode](#)
 - Parameter [AutomaticBusOffRecovery](#)
 - Parameter [ManualBusOffRecovery](#)
 - Reference [CanControllerDefaultBaudrate](#)
 - Reference [CanControllerEcucPartitionRef](#)
 - Reference [CanCpuClockRef](#)
 - Reference [CanWakeupSourceRef](#)
 - Container [CanControllerBaudrateConfig](#)
 - Parameter [CanAdvancedSetting](#)
 - Parameter [CanBusLength](#)

- Parameter [CanPropDelayTranceiver](#)
- Parameter [CanControllerCheckCanStandard](#)
- Parameter [CanControllerPrescaler](#)
- Parameter [CanControllerBaudRateConfigID](#)
- Parameter [CanControllerBaudRate](#)
- Parameter [CanControllerSyncSeg](#)
- Parameter [CanControllerPropSeg](#)
- Parameter [CanControllerSeg1](#)
- Parameter [CanControllerSeg2](#)
- Parameter [CanControllerSyncJumpWidth](#)
- Container [CanControllerFdBaudrateConfig](#)
- Parameter [CanAdvancedSettingFd](#)
- Parameter [CanBusLengthFd](#)
- Parameter [CanControllerFDPrescaler](#)
- Parameter [CanControllerFdBaudRate](#)
- Parameter [CanControllerFdSyncSeg](#)
- Parameter [CanControllerPropSeg](#)
- Parameter [CanControllerSeg1](#)
- Parameter [CanControllerSeg2](#)
- Parameter [CanControllerSyncJumpWidth](#)
- Parameter [CanControllerSspOffset](#)
- Parameter [CanControllerTxBitRateSwitch](#)
- Container [CanTTController](#)
- Parameter [CanTTControllerApplWatchdogLimit](#)
- Parameter [CanTTControllerCycleCountMax](#)
- Parameter [CanTTControllerExpectedTxTrigger](#)
- Parameter [CanTTControllerExternalClockSynchronisation](#)
- Parameter [CanTTControllerGlobalTimeFiltering](#)
- Parameter [CanTTControllerInitialRefOffset](#)
- Parameter [CanTTControllerInterruptEnable](#)
- Parameter [CanTTControllerLevel2](#)
- Parameter [CanTTControllerNTUConfig](#)
- Parameter [CanTTControllerOperationMode](#)
- Parameter [CanTTControllerSyncDeviation](#)
- Parameter [CanTTControllerTURRestore](#)
- Parameter [CanTTControllerTimeMaster](#)
- Parameter [CanTTControllerTimeMasterPriority](#)
- Parameter [CanTTControllerTxEnableWindowLength](#)
- Parameter [CanTTControllerWatchTriggerGapTimeMark](#)
- Parameter [CanTTControllerWatchTriggerTimeMark](#)
- Parameter [CanTTIRQProcessing](#)
- Reference [CanTTControllerEcucPartitionRef](#)
- * Container [CanHardwareObject](#)
 - Parameter [Dummy_HRH](#)
 - Parameter [CanFdPaddingValue](#)
 - Parameter [CanHandleType](#)

- Parameter [CanIdType](#)
- Parameter [CanObjectId](#)
- Parameter [CanObjectType](#)
- Parameter [CanTxAddFrameMac](#)
- Parameter [CanLlceKeyHandle](#)
- Parameter [CanHardwareObjectUsesPolling](#)
- Parameter [CanTriggerTransmitEnable](#)
- Parameter [CanHwObjectCount](#)
- Parameter [CanObjectPayloadLength](#)
- Reference [CanControllerRef](#)
- Reference [CanMainFunctionRWPeriodRef](#)
- Container [CanHwFilter](#)
- Parameter [CanHwFilterCode](#)
- Parameter [CanHwFilterMask](#)
- Container [RangeFilter](#)
- Parameter [RangeStart](#)
- Parameter [RangeEnd](#)
- Container [CanAdvancedFeature](#)
- Reference [CanAdvancedFeatureRef](#)
- Container [CanTTHardwareObjectTrigger](#)
- Parameter [CanTTHardwareObjectBaseCycle](#)
- Parameter [CanTTHardwareObjectCycleRepetition](#)
- Parameter [CanTTHardwareObjectTimeMark](#)
- Parameter [CanTTHardwareObjectTriggerId](#)
- Parameter [CanTTHardwareObjectTriggerType](#)
- * Container [CanAuxFilter](#)
 - Parameter [CanIdType](#)
 - Parameter [Filter_Type](#)
 - Parameter [AuxSearchResult](#)
 - Parameter [CanHwFilterCode](#)
 - Parameter [CanHwFilterMask](#)
 - Parameter [RangeStart](#)
 - Parameter [RangeEnd](#)
 - Reference [CanControllerRef](#)
- * Container [CanIcom](#)

- Container [CanIcomConfig](#)
- Parameter [CanIcomConfigId](#)
- Parameter [CanIcomWakeOnBusOff](#)
- Container [CanIcomWakeupCauses](#)
- Container [CanIcomRxMessage](#)
- Parameter [CanIcomCounterValue](#)
- Parameter [CanIcomMessageId](#)
- Parameter [CanIcomIdOperation](#)
- Parameter [CanIcomMessageIdMask](#)
- Parameter [CanIcomMissingMessageTimerValue](#)
- Parameter [CanIcomPayloadLengthError](#)
- Container [CanIcomRxMessageSignalConfig](#)
- Parameter [CanIcomSignalMask](#)
- Parameter [CanIcomSignalOperation](#)
- Parameter [CanIcomSignalValue](#)
- Parameter [DLCLowValue](#)
- Parameter [DLCHighValue](#)
- Reference [CanIcomSignalRef](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)

4.1 Module Can

This container holds the configuration of a single CAN Driver.

Included containers:

- [CanGeneral](#)
- [CanConfigSet](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container CanGeneral

This container holds the parameters related each CAN Driver Unit.

Included subcontainers:

- [CanMainFunctionRWPeriods](#)
- [CanIcomGeneral](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter CanDevErrorDetect

CAN064_Conf: Switches the Development Error Detection and Notification: ON or OFF.

When this option is OFF code size is reduced, but no error detection is available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.4 Parameter CanEnableUserModeSupport

When this parameter is enabled, the Can module will adapt to run from User Mode

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.5 Parameter CanMulticoreSupport

Enable mapping Can LLCE driver to multiple EcuC partitions to make the modules API available in this partition. The Can driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.6 Parameter HostInterface

Specifies the selection for the host interface

NoteImplementation Specific parameter. Not AutoSar Required.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	LLCE_CAN_HIF0
literals	['LLCE_CAN_HIF0', 'LLCE_CAN_HIF1']

4.7 Parameter CanVersionInfoApi

CAN106_Conf. Switches the Can_GetVersionInfo() API: ON or OFF.

When this option is ON driver supports API for getting Version information for the Driver.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.8 Parameter CanIndex

CAN320_Conf. Specifies the InstanceId of this module instance.

If only one instance is present it shall have the Id 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	255
min	0

4.9 Parameter CanMainFunctionBusoffPeriod

CAN355_Conf. This parameter describes the period for cyclic call to Can_MainFunction_Busoff.

Unit is seconds.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0.001
max	65.535
min	0.001

4.10 Parameter CanMainFunctionWakeupPeriod

CAN357_Conf. This parameter describes the period for cyclic call to Can_MainFunction_Wakeup.

Unit is seconds.

This field is editable if CanConfigSet/CanController/CanWakeupSupport is 'true'.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0.001
max	65.535
min	0.001

4.11 Parameter CanMainFunctionModePeriod

CAN376_Conf. This parameter describes the period for cyclic call to Can_MainFunction_Mode.

Unit in seconds.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0.001
max	65.535
min	0.001

4.12 Parameter CanMultiplexedTransmission

Multiplex transmission means to search for a free MB, that has the same ObjectId with the one transmitted to Can_Write,

if current Hth MB is busy.

Replaced by SWS_Can_00401. There is no difference between configuring multiple transmit

hardware handlers with the same CanObjectId and configuring a single transmit hardware handler with multiple transmit

hardware objects. Hence, it is advisable to use the CanHwObjectCount mechanism to allocate more free transmit hardware objects within a transmit entity.

Note: Deprecated. Please use CanHwObjectCount functionality.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.13 Parameter CanTimeoutMethod

CanTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If SystemTimer or CustomTimer are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_DUMMY', 'OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM']

4.14 Parameter CanTimeoutDuration

CAN113_Conf. Specifies the maximum time for blocking function until a timeout is detected. Unit in seconds.

This Timeout is used to detect the Hardware Errors/ Production Errors.

This timeout is used to allow hardware to take effect of the Hardware settings.

For OSIF_COUNTER_DUMMY method, CanTimeoutDuration may not reflect exactly in seconds (but in the terms of loops, 1 us : 1 loop).

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1.0
max	65.535
min	1.0E-6

4.15 Parameter CanLPduReceiveCalloutFunction

CAN434_Conf: This parameter defines the existence and the name of a callout function that is called after a successful reception of a received CAN Rx L-PDU. If this parameter is omitted no callout shall take place.

Property	Value
type	ECUC-FUNCTION-NAME-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.16 Parameter LPDUCalloutExtension

Allows LLCE callout Extension to accept 2 bytes unsigned integer values.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.17 Parameter RxTimestampNotification

Vendor specific: Timestamp of received frames are reported through this callout function, if any controller has timestamp option enabled.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	NULL_PTR

4.18 Parameter TxTimestampNotification

Vendor specific: Timestamp of frames confirmations are reported through this callout function, if any controller has timestamp option enabled.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	NULL_PTR

4.19 Parameter CanSetBaudrateApi

If the parameter is set to true the Can_SetBaudrate Api shall be supported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.20 Parameter CanApiEnableMbAbort

Enables an additional API which request to abort the lowest priority pending transmission belonging to a channel

Based on this value a define will be generated with STD_ON or STD_OFF value: CAN_43_LLCE_API_ENABLE_ABORT

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.21 Parameter CanAbortOnlyOneMB

When this parameter is true only one MB will be aborted by Can_AbortMb()

When this parameter is false all the associated message buffers will be aborted for a specific HTH

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.22 Parameter CanLlceEnableHeadlessMode

Set if LLCE Headless mode will be used in any variant

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.23 Parameter CanLlceHeadlessModeUsed

Set if LLCE is pre-loaded with a headless configuration for this variant

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.24 Parameter CanCustomRxFunction

Defines the existence and the name of a callback function that is called after a successful reception of a CAN frame received through a custom filter.

Leave disabled to stop callbacks in driver.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.25 Parameter CanWriteCustomCallback

Defines the existence and the name of a callback function that is called for each frame pushed to LLCE firmware.

It provides the HoH, swPduHandle and Message Buffer Index of the transmitted frame. When the callback is triggered,

the message is already written into the buffer based on the Index value.

Leave disabled to stop the callback in driver.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.26 Parameter CanTxConfirmationCustomCallback

Defines the existence and the name of a custom callback function that is called for each Tx Confirmation received from LLCE firmware.

It provides the HoH and swPduHandle.

Leave disabled to stop the callback in driver.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.27 Parameter CanAuxFilterEnable

This checkbox enables the CanAuxFilter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.28 Parameter CanAuxFilterCallback

Defines the existence and the name of a custom callback function that is called for RxLut2 result processing.

Leave disabled to stop the callback in driver.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	NULL_PTR

4.29 Parameter BusOffNotification

Vendor specific: When the driver is configured in bus-off recovery (automatic or manual),

the LLCE_ERROR_BUSOFF notification triggered by LLCE firmware will call this function, providing the u8CanIfCtrlId.

In case of Manual Bus Off Recovery, the user may get the controller out of bus-off confinement state

using the API Can_43_LLCE_ManualBusOffRecovery(uint8 u8Controller).

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.30 Parameter CanPublicIcomSupport

Selects support of Pretended Network features in Can driver. Fixed to false (Pretended Networking not supported)

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.31 Reference CanEcucPartitionRef

ECUC_Can_00491. Maps the CAN driver to zero or multiple ECUC partitions to make the modules API available in this partition. The CAN driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.32 Reference CanOsCounterRef

CAN431_Conf. This parameter contains a reference to the counter, which is used by the CAN driver.

Note: This node is unused, the using of OsCounter is done by selecting CanTimeoutMethod to OSIF_COUNTER_SYST

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Os/OsCounter

4.33 Reference CanSupportTTCANRef

CAN430_Conf. Reference to CAN Hardware Object which shall be polled with the configured CanMainFunctionWritePeriod.

This reference shall only be configurable if more than one period is configured via CanMainFunctionWritePeriod.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/CanIf/CanIfPrivateCfg

4.34 Container CanMainFunctionRWPeriods

ECUC_Can_00437. This container contains the parameter for configuring the period for cyclic call to Can_MainFunction_Read or Can_MainFunction_Write depending on the referring item.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.35 Parameter CanMainFunctionPeriod

ECUC_Can_00484. This parameter describes the period for cyclic call to Can_MainFunction_Read or Can_MainFunction_Write depending on the referring item. Unit is seconds.

Different poll-cycles will be configurable if more than one CanMainFunctionPeriod is configured.

In this case multiple Can_MainFunction_Read() or Can_MainFunction_Write() will be provided by the CAN Driver module.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.001
max	65.535
min	0.001

4.36 Parameter AckInterface

This parameter specifies the number of order of ACK interface for the current polling class.

First 16 interfaces are reserved for interrupt rx processing.

In case of multihost usecase, the user should take care to not assign same interfaces on different hosts.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	17
max	21
min	16

4.37 Parameter RxInterface

This parameter specifies the number of order of Rx interface for the current polling class.

First 16 interfaces are reserved for interrupt ack processing.

In case of multihost usecase, the user should take care to not assign same interfaces on different hosts.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	17
max	21
min	16

4.38 Container CanIcomGeneral

This container contains the general configuration parameters of the ICOM Configuration

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.39 Parameter CanIcomLevel

Defines the level of Pretended Networking. This parameter is reserved for future implementations (Pretended Networking level 2).

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	CAN_ICOM_LEVEL_ONE
literals	['CAN_ICOM_LEVEL_ONE', 'CAN_ICOM_LEVEL_TWO']

4.40 Parameter CanIcomVariant

Defines the variant, which is supported by this CanController

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	CAN_ICOM_VARIANT_NONE
literals	['CAN_ICOM_VARIANT_HW', 'CAN_ICOM_VARIANT_NONE', 'CAN_ICOM_VARIANT_SW']

4.41 Container CanConfigSet

CAN343_Conf. This is the multiple configuration set container for CAN Driver.

Included subcontainers:

- [CanErrorReporting](#)
- [CanController](#)
- [CanHardwareObject](#)
- [CanAuxFilter](#)
- [CanIcom](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.42 Container CanErrorReporting

This container includes the possible errors that CAN_43_LLCE plugin can issue.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.43 Parameter CAN__PROTOCOL__ERR

This is recommended to be enabled only for debugging purposes, since it may generate very frequent notifications.

This category includes the following errors:

BIT0ERR This bit indicates when an inconsistency occurs between the transmitted and the received bit in a CAN frame

BIT1ERR: This bit indicates when an inconsistency occurs between the transmitted and the received bit in a CAN frame

ACKERR This bit indicates that an Acknowledge Error has been detected by the transmitter node, that is, a dominant bit has not been detected during the ACK SLOT.

CRCERR: This bit indicates that a CRC Error has been detected by the receiver node in a CAN frame, that is, the calculated CRC is different from the received.

STFERR: This bit indicates that a Stuffing Error has been detected by the receiver node in a CAN frame.

FRMERR: This bit indicates that a Form Error has been detected by the receiver node in a CAN frame, that is, a fixed-form bit field contains at least one illegal bit

OTHER_BCAN_ERR:

ERROVR(Error Overrun bit)- This bit indicates that an error condition occurred when any error bit is already set.

DPCRCERR(Data phase Cyclic Redundancy Check Error)- This bit indicates that a CRC Error has been detected by the receiver node in the data phase

of a CAN FD frame, that is, the calculated CRC is different from the received.

DPFRMERR(Data phase form error) - This bit indicates that a Form Error has been detected by the receiver node in the data phase

of a CAN FD frame, that is, a fixed-form bit field contains at least one illegal bit.

DPSTFERR(Data phase stuff error) - This bit indicates that a Stuffing Error has been detected by the receiver node in the data

phase of a CAN FD frame

DPBIT0ERR(Data phase bit0 error) - This bit indicates when an inconsistency occurs between the transmitted and the received bit

in the data phase of a CAN FD frame.

DPBIT1ERR(Data phase bit1 error) - This bit indicates when an inconsistency occurs between the transmitted and the received bit

in the data phase of a CAN FD frame

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	IGNORE
literals	['IGNORE', 'INTERRUPT', 'POLLING']

4.44 Parameter CAN_PROTOCOL_WRN

This category includes the following warnings:

- Passive error state. PASSERR is set when BCAN enters Passive state.
- Tx error warnings. TXWRN is set when the Tx error counter ECR[TEC] reached 96.
- Rx error warnings. RXWRN is set when the Rx error counter ECR[REC] reached 96.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	INTERRUPT
literals	['IGNORE', 'INTERRUPT', 'POLLING']

4.45 Parameter DATA_LOST_ERR

DATA_LOST event as described by Autosar standard.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	INTERRUPT
literals	['IGNORE', 'INTERRUPT', 'POLLING']

4.46 Parameter CONFIGURATION_ERR

Configuration errors, like resources overflow, bad commands flow, bad command parameters, invalid indexes or any other errors caused by bad usage or a malicious host.

Default set is 'IGNORE', because the same error codes are sent to host through shared memory area of each command.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	IGNORE
literals	['IGNORE', 'INTERRUPT', 'POLLING']

4.47 Parameter INTERNAL_ERR

Internal errors, like timeouts, empty/full fifos or other software errors.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	INTERRUPT
literals	['IGNORE', 'INTERRUPT', 'POLLING']

4.48 Parameter CanErrorNotif

Vendor specific: The errors triggered by LLCE firmware will call this function.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	NULL_PTR

4.49 Parameter CanMainFunctionErrorEn

Enables/Disables MainFunction_ErrorNotification API for reporting the errors configured in POLLING.

NoteImplementation Specific parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.50 Container CanController

CAN354_Conf. This container contains the configuration parameters of the CAN controller(s).

Included subcontainers:

- [CanControllerBaudrateConfig](#)
- [CanTTCController](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE

4.51 Parameter CanHwChannel

Specifies which one of the on-chip interfaces is associated with this controller ID.

NoteImplementation Specific parameter. Not AutoSar Required.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	BCAN_0
literals	['BCAN_0', 'BCAN_1', 'BCAN_2', 'BCAN_3', 'BCAN_4', 'BCAN_5', 'BCAN_6', 'BCAN_7', 'BCAN_8', 'BCAN_9', 'BCAN_10', 'BCAN_11', 'BCAN_12', 'BCAN_13', 'BCAN_14', 'BCAN_15']

4.52 Parameter RWDestInterface

FIFO Interface ID used for interrupt processing:

1. to send the frames received by a filter configured for CanHwChannel (LLCE_FIFO_RXOUT_BASE_ADDRESS(x)).
2. to acknowledge the frames sent by CanHwChannel (LLCE_FIFO_TXACK_BASE_ADDRESS(x)).

Default value is 1 to 1 mapped with Can Hardware Channel.

NoteUsage of this field may violate AutoSar standard.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FIFO_INTERFACE_0
literals	['FIFO_INTERFACE_0', 'FIFO_INTERFACE_1', 'FIFO_INTERFACE_2', 'FIFO_INTERFACE_3', 'FIFO_INTERFACE_4', 'FIFO_INTERFACE_5', 'FIFO_INTERFACE_6', 'FIFO_INTERFACE_7', 'FIFO_INTERFACE_8', 'FIFO_INTERFACE_9', 'FIFO_INTERFACE_10', 'FIFO_INTERFACE_11', 'FIFO_INTERFACE_12', 'FIFO_INTERFACE_13', 'FIFO_INTERFACE_14', 'FIFO_INTERFACE_15']

4.53 Parameter CanControllerActivation

CAN315_Conf. Defines if a CAN controller is used in the configuration.

Deactivation of a particular CAN controller is equivalent to a CAN controller not used in the configuration.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.54 Parameter CanControllerBaseAddress

CAN382_Conf. Specifies the CAN controller base address.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	4294967295
min	0

4.55 Parameter CanControllerIcomSupported

Define if the current controller support Pretended Networking (Icom)

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. Fixed to false

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.56 Parameter CanControllerId

CAN316_Conf: This parameter provides the controller ID which is unique in a given CAN Driver.

The value for this parameter starts with 0 and continue without any gaps.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1
max	255
min	0

4.57 Parameter CanRxProcessing

CAN317_Conf. Enables/Disables API Can_MainFunction_Read() for handling PDU reception events in polling mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	POLLING
literals	['INTERRUPT', 'POLLING', 'MIXED']

4.58 Parameter CanTxProcessing

CAN318_Conf. Enables/Disables API Can_MainFunction_Write() for handling PDU transmission events in polling mode

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	POLLING
literals	['INTERRUPT', 'POLLING', 'MIXED']

4.59 Parameter CanBusoffProcessing

CAN314_Conf. Enables/Disables API Can_MainFunction_BusOff() for handling busoff events in polling mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	POLLING
literals	['INTERRUPT', 'POLLING']

4.60 Parameter CanWakeupFunctionalityAPI

Adds / removes the service Can_CheckWakeup() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.61 Parameter CanWakeupProcessing

CAN319_Conf. Enables/Disables API Can_MainFunction_Wakeup() for handling wakeup events in polling mode.

Note This option is enabled only if global parameter [CanController/CanWakeupSupport] is 'true'.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	POLLING
literals	['INTERRUPT', 'POLLING']

4.62 Parameter CanWakeupSupport

CAN330_Conf. CAN driver support for wakeup over CAN Bus.

Every WakeUp process will be ignore if this checkbox is not set to ON.

This parameter enables Internal Wakeup (using controller registers) and External Wakeup (using WKUP module).

This is enabled only if internal Wakeup is supported by the platform.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.63 Parameter CanTimeStampEnable

Enables CAN to put timestamp on frames.

NoteImplementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DISABLE
literals	['DISABLE', 'END_OF_FRAME', 'START_OF_FRAME', 'CAN_START_↵ OF_FRAME_CANFD_RES']

4.64 Parameter CanProtocolException

Enable Protocol Exception feature in the BCAN controller during initialization.

Enable of this feature will allow that BCAN controller to set

an interrupt status flag whenever a specific protocol error is detected.

NoteImplementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.65 Parameter CanListenOnlyMode

Enables the Listen Only Mode (CTRL[LOM] bit).

In this mode, transmission is disabled, all error counters are frozen and the module operates in a CAN Error Passive mode.

NoteImplementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.66 Parameter CanLoopBackMode

CTRL[LPB]: Enables CAN to operate in Loop Back Mode.

The bit stream output of the transmitter is fed back internally to the receiver input.

The Rx CAN input pin is ignored and the Tx CAN output goes to the recessive state (logic "1").

NoteImplementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.67 Parameter SelfReceptionEnable

CTRL[SRXEN]: Enables CAN to receive frames transmitted by itself.

This bit purpose is for debug, it is recommended keeping it disabled to avoid overload.

NoteImplementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.68 Parameter CanTxFifoMode

This field defines how the TX messages are processed: in a FIFO manner or based on the bus arbitration algorithm.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.69 Parameter AutomaticBusOffRecovery

This field defines how BCAN recovers from bus-off state. If this field is checked, the module recovers automatically from bus-off state after 128 occurrences of 11 recessive bits. If the bit is unchecked, automatic recovery from bus-off is disabled and the module remains in bus-off state until the BCAN is started by the user (As per AUTOSAR) and 128 occurrences of 11 recessive bits have occurred.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.70 Parameter ManualBusOffRecovery

A new API is exposed: Can_43_LLCE_ManualBusOffRecovery(uint8 u8Controller).

Enabling this setting overwrites the Autosar requirement for busoff: it does not set the controller to STOP and it does not call the CanIf callout. Instead, a user defined callout shall be defined as BusOffNotification.

This field defines how BCAN recovers from bus-off state. If this field is checked, the module remains in bus-off state until the user manually requests the controller to recover from bus-off state without flushing the outstanding TX or RX frames. The controller is considered started and LLCE may accept and store new transmissions, which will be serviced as soon as the offending condition is removed and the user calls the exposed API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.71 Reference CanControllerDefaultBaudrate

CAN435_Conf. Reference to baudrate configuration container configured for the Can Controller.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Can/CanConfigSet/CanController/CanController↔ BaudrateConfig

4.72 Reference CanControllerEcucPartitionRef

ECUC_Can_00492. Maps the CAN controller to zero or one ECUC partitions. The ECUC

partition referenced is a subset of the ECUC partitions where the CAN

driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true

Property	Value
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.73 Reference CanCpuClockRef

CAN313_Conf. Reference to the CPU clock configuration, which is set in the MCU driver configuration.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.74 Reference CanWakeupSourceRef

CAN359_Conf. This parameter contains a reference to the Wakeup Source for this controller as defined in the ECU State Manager.

Type: reference to EcuM_WakeupSourceType

EcuM plugin need to be added and then give the reference to it.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuM/EcuMConfiguration/EcuMCommon← Configuration/EcuMWakeupSource

4.75 Container CanControllerBaudrateConfig

CAN351. This container contains the configuration (parameters) of the CAN Filter Mask(s).

Included subcontainers:

- [CanControllerFdBaudrateConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.76 Parameter CanAdvancedSetting

If TRUE initiates the derivation of the CAN bit timing values from the CanControllerBaudRate parameter.

When this option is True the CanControllerPropSeg, CanControllerSeg1, CanControllerSeg2, CanControllerSyncJumpWidth are disabled because are background calculated.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.77 Parameter CanBusLength

Specifies the CAN Bus length in meters.

This parameter is used for PROPSEG parameter calculation when "CanAdvancedSetting" control is set to true.

NoteImplementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	40
max	5000
min	1

4.78 Parameter CanPropDelayTranceiver

Propagation delay of Transceiver used in nanoseconds.

NoteImplementation specific Parameter.

The calculation for the CAN bit timing is implemented in the code template.

The Formulas used in the code template for calculation are as follows.

Physical delay of bus = Bus length * Bus propagation delay.

$t_{PROP_SEG} = 2(\text{Physical delay of bus} + \text{CanPropDelayTranceiver})$.

$PROP_SEG = \text{ROUND_UP}(t_{PROP_SEG} / \text{Bus propagation delay})$.

Based on these calculations implemented in the Code template the consistency check is maintained for CanPropDelayTranceiver parameter.

The PROP_SEG parameter need to be a integral value and not fractional value.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	150.0
max	5000.0
min	0.0

4.79 Parameter CanControllerCheckCanStandard

Note

EN: If enabled, Can Plugin checks that CanControllerPropSeg, CanControllerSeg1, CanControllerSeg2 and CanControllerSyncJump settings

match the CAN Standard Compliant Bit Time Segment Settings.

See the following table for details:

CanControllerPropSeg + CanControllerSeg1

CanControllerSeg2

CanControllerSyncJumpWidth

[5 .. 10] 2 [1 .. 2]

[4 .. 11] 3 [1 .. 3]

[5 .. 12] 4 [1 .. 4]

[6 .. 13] 5 [1 .. 4]

[7 .. 14] 6 [1 .. 4]

[8 .. 15] 7 [1 .. 4]

[9 .. 16] 8 [1 .. 4]

NoteImplementation Specific parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.80 Parameter CanControllerPrescaler

Specifies the prescaler for the controller .

The calculation of the resulting CanControllerTimeQuanta value depending on module clocking and prescaler shall be done offline.

Prescaler = FreqCanClk / FreqTq; FreqTq = 1 / CanControllerTimeQuanta .

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	16
max	1023
min	1

4.81 Parameter CanControllerBaudRateConfigID

Uniquely identifies a specific baud rate configuration. This ID is used by

SetBaudrate API

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65535
min	0

4.82 Parameter CanControllerBaudRate

CAN005_Conf. Specifies the buadrate of the controller in kbps.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	147.05882352941177
max	1000.0
min	0.0

4.83 Parameter CanControllerSyncSeg

The Synchronization Segment or SYNC_SEG time interval is used to synchronize all the nodes across the network.

The SYNC_SEG time interval has a fixed period of one Time Quantum (TQ).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	1
min	1

4.84 Parameter CanControllerPropSeg

CAN073_Conf. It is used to compensate the physical delay within the CAN network (CTRL[PROPSEG] - 0..7).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	5
max	255
min	0

4.85 Parameter CanControllerSeg1

CAN074_Conf. Specifies the Phase Segment 1 in time quantas.

$$\text{PHASE_BUF_SEG1} = (\text{PSEG1} + 1) * T_q$$

The PHASE_BUF_SEG1 valid values are 2-256 T_q.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	5
max	255
min	2

4.86 Parameter CanControllerSeg2

The PHASE_BUF_SEG2 valid values are 2-128 T_q.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	6
max	128
min	2

4.87 Parameter CanControllerSyncJumpWidth

NoteImplementation specific Parameter. Not AutoSar required.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	127
min	0

4.88 Container CanControllerFdBaudrateConfig

This optional container contains bit timing related configuration parameters of the CAN controller(s) for payload and CRC of a CAN FD frame. If this container exists the controller supports CAN FD frames.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.89 Parameter CanAdvancedSettingFd

If TRUE initiates the derivation of the CAN bit timing values from the CanControllerBaudRate parameter.

When this option is True the CanControllerPropSeg, CanControllerSeg1, CanControllerSeg2, CanControllerSyncJumpWidth are disabled because are background calculated.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.90 Parameter CanBusLengthFd

Specifies the CAN Bus length in meters.

This parameter is used for PROPSEG parameter calculation when "CanAdvancedSettingFd" control is set to true.

NoteImplementation specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	40
max	5000
min	1

4.91 Parameter CanControllerFDPrescaller

Specifies the prescaller for the controller in FD mode .

The calculation of the resulting CanControllerTimeQuanta value depending on module clocking and prescaler shall be done offline.

$\text{Prescaler} = \text{FreqCanClk} / \text{FreqTq}; \text{FreqTq} = 1 / \text{CanControllerTimeQuanta} .$

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	16
max	63
min	1

4.92 Parameter CanControllerFdBaudRate

ECUC_Can_00481.Specifies the data segment baud rate of the controller in kbps.

CAN FD maximum speed is 8Mbps.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	8000.0
max	16000.0
min	0.0

4.93 Parameter CanControllerFdSyncSeg

The Synchronization Segment or SYNC_SEG time interval is used to synchronize all the nodes across the network.

The SYNC_SEG time interval has a fixed period of one Time Quantum (TQ).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	1
min	1

4.94 Parameter CanControllerPropSeg

ECUC_Can_00476.Specifies propagation delay in time quantas.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	255
min	0

4.95 Parameter CanControllerSeg1

ECUC_Can_00477.Specifies phase segment 1 in time quantas.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	255
min	0

4.96 Parameter CanControllerSeg2

ECUC_Can_00478.Specifies phase segment 2 in time quantas.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	6
max	255
min	0

4.97 Parameter CanControllerSyncJumpWidth

ECUC_Can_00479.Specifies the synchronization jump width for the controller in time quantas.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	15
min	0

4.98 Parameter CanControllerSspOffset

ECUC_Can_00494. Specifies the Transmitter Delay Compensation Offset in minimum time quanta

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	255
min	0

4.99 Parameter CanControllerTxBitRateSwitch

Specifies if the bit rate switching shall be used for transmissions.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.100 Container CanTTController

TTCan IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.101 Parameter CanTTControllerApplWatchdogLimit

Defines the maximum time period (unit is 256 times NTU) after which the application has to serve the watchdog.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.102 Parameter CanTTControllerCycleCountMax

Defines the value for cycle_count_max.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	63
min	0

4.103 Parameter CanTTControllerExpectedTxTrigger

Number of expected_tx_trigger.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	255
min	0

4.104 Parameter CanTTControllerExternalClockSynchronisation

Enables/disables the external clock synchronization.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.105 Parameter CanTTControllerGlobalTimeFiltering

Enables/disables the global time filtering.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.106 Parameter CanTTControllerInitialRefOffset

Defines the initial value for ref trigger offset.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	127
min	0

4.107 Parameter CanTTControllerInterruptEnable

Enables/disables the respective interrupts.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	1023
min	0

4.108 Parameter CanTTControllerLevel2

Defines whether Level 2 or Level 1 is used.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.109 Parameter CanTTControllerNTUConfig

Defines the config value for NTU (network time unit).

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0.0
max	100.0
min	0.0

4.110 Parameter CanTTControllerOperationMode

Defines the operation mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	CAN_TT_EVENT_SYNC_TIME_TRIGGERED
literals	['CAN_TT_EVENT_SYNC_TIME_TRIGGERED', 'CAN_TT_EVENT_↵ TRIGGERED', 'CAN_TT_TIME_TRIGGERED']

4.111 Parameter CanTTControllerSyncDeviation

Defines the maximum synchronization deviation:

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	100.0
min	0.0

4.112 Parameter CanTTControllerTURRestore

Enables/disables the TUR restore.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.113 Parameter CanTTControllerTimeMaster

Defines whether the controller acts as a potential time master.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.114 Parameter CanTTControllerTimeMasterPriority

Defines the time master priority.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	7
min	0

4.115 Parameter CanTTControllerTxEnableWindowLength

Length of the tx enable window given in CAN bit times.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	1
max	16
min	1

4.116 Parameter CanTTControllerWatchTriggerGapTimeMark

watch trigger time mark after a gap

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0

Property	Value
max	65535
min	0

4.117 Parameter CanTTControllerWatchTriggerTimeMark

watch trigger time mark

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	65535
min	0

4.118 Parameter CanTTIRQProcessing

Enables / disables API Can_MainFunction_BusOff() for handling busoff events in polling mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	POLLING
literals	['INTERRUPT', 'POLLING']

4.119 Reference CanTTControllerEcucPartitionRef

ECUC_Can_00493. Maps the Time triggered CAN controller to zero or one ECUC

partitions. The ECUC partition referenced is a subset of the ECUC

partitions where the CAN driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.120 Container CanHardwareObject

CAN_323. This container contains the configuration (parameters) of CAN Hardware Objects.

This configuration element is used as information for the CAN Interface only.

The relevant CAN driver configuration is done with the filter mask and identifier.

Included subcontainers:

- [CanHwFilter](#)
- [RangeFilter](#)
- [CanAdvancedFeature](#)
- [CanTTHardwareObjectTrigger](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

S32G CAN-LLCE Driver

4.121 Parameter Dummy_HRH

Marks the HRH as dummy. It acts as a placeholder in order to enable setting a filter without the need of removing another one

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.122 Parameter CanFdPaddingValue

MBCS[PRI0]: This value it is the padding value when FD it is used.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	255
min	0

4.123 Parameter CanHandleType

CAN323_Conf. Specifies the type (Full-CAN or Basic-CAN) of a hardware object.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	BASIC
literals	['BASIC', 'FULL']

4.124 Parameter CanIdType

CAN065_Conf. Specifies whether the IdValue is of type

- standard identifier (ID - 11 bits length)
- extended identifier (ID - 29 bits length)
- mixed mode (standard or extended)

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	STANDARD
literals	['EXTENDED', 'MIXED', 'STANDARD']

4.125 Parameter CanObjectId

CAN326_Conf. Holds the handle ID of HRH or HTH.

The value of this parameter is unique in a given CAN Driver, and it should start with 0 and continue without any gaps.

The HRH and HTH Ids are defined under two different name-spaces.

Example: HRH0-0, HRH1-1, HTH0-2, HTH1-3

Property	Value
type	ECUC-INTEGGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65535
min	0

4.126 Parameter CanObjectType

CAN327_Conf. Specifies if the HardwareObject is used as Transmit or as Receive object.

NoteMBs configured as MIXED standard and RECEIVE type will be treated as EXTENDED.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	RECEIVE
literals	['RECEIVE', 'TRANSMIT']

4.127 Parameter CanTxAddFrameMac

Enable adding of MAC frames to the frames trasmitted by this HTH.

Enable of this feature allows changing the payload size by inserting at the end the generated MAC code.

Used only for HTHs.

Note Implementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.128 Parameter CanLlceKeyHandle

Key handle for MAC generation.

Used only for HTHs.

Note Implementation specific Parameter. Not AutoSar Required.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	S32G CAN_LLCE Driver

4.129 Parameter CanHardwareObjectUsesPolling

Enables polling of this hardware object. This node shall exist if CanRxProcessing/CanTxProcessing is set to Mixed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.130 Parameter CanTriggerTransmitEnable

This parameter defines if or if not Can supports the trigger-transmit API for this handle.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.131 Parameter CanHwObjectCount

Number of hardware objects used to implement one HOH. In case of a HRH this parameter defines the number of elements in the hardware FIFO or the number of shadow buffers, in case of a HTH it defines the number of hardware objects used for multiplexed transmission or for a hardware FIFO used by a FullCAN HTH

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	1
max	2397
min	1

4.132 Parameter CanObjectPayloadLength

EN: Enabling this checkbox makes the HwObjects be available only for short CAN/CAN-FD frames (8-bytes frames)

instead of the default long CAN/CAN-FD frames (64-bytes frames). There are 2016 long HwObjects and 2080 short HwObjects available for reception.

Note: This works only for RECEIVE Objects. Only PL_8 and PL_64 work , the rest of them are treated as pl_64

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	CAN_OBJECT_PL_64
literals	['CAN_OBJECT_PL_8', 'CAN_OBJECT_PL_12', 'CAN_OBJECT_PL_16', 'CAN_OBJECT_PL_20', 'CAN_OBJECT_PL_24', 'CAN_OBJECT_PL_32', 'CAN_OBJECT_PL_48', 'CAN_OBJECT_PL_64']

4.133 Reference CanControllerRef

CAN322_Conf. Reference to CAN Controller to which the HOH is associated to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Can/CanConfigSet/CanController

4.134 Reference CanMainFunctionRWPeriodRef

CAN438_Conf.Reference to CAN Controller to which the HOH is associated to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Can/CanGeneral/CanMainFunctionRWPeriods

4.135 Container CanHwFilter

ECUC_Can_00468 : This container is only valid for HRHs and contains the configuration (parameters) of one hardware filter.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.136 Parameter CanHwFilterCode

CAN325_Conf. Specifies (together with the filter mask)- the identifiers range that passes the hardware filter for of RX objects.

Parameter ranges from 0 to 0x7FF (11 bits) for Standard IDs and 0 to 0xFFFFFFFF (29 bits) for Extended IDs.

User can assign any code to this parameter, but must to respect the above rule related to Standard/Extended IDs.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	0

4.137 Parameter CanHwFilterMask

ECUC_Can_00469 : Specifies (together with the filter mask) the identifiers range that passes the hardware filter.

Different CanHardwareObjects with different CanIdTypes (STANDARD, MIXED, EXTENDED)

can share the same CanFilterMask (i.e., the CanFilterMaskRef parameters of these

CanHardwareObjects reference the very same CanFilterMask container). This shall be allowed and must be supported by the configuration generators.

The CanFilterMaskRef is omitted for

- 1) CanHardwareObjects with CanObjectType set to TRANSMIT
- 2) CanHardwareObjects with CanObjectType set to RECEIVE if only a single Can ID

shall be received via this CanHardwareObjects(i.e. exact match with CanIdValue requested when CanHandleType is configured as FULL)

CAN321_Conf. Reference to the filter mask that is used for hardware filtering together with the CAN_ID_VALUE.

This value is used as acceptance masks for ID filtering in RX MBs.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	0

4.138 Container RangeFilter

Specify that this filter is of range type. This over-rides the information in the standard CanHwFilter.

Being a non Autosar concept, the range filter discards the information of CanHandleType and CanIdType.

If enabled, the filter will accept IDs from RangeStart to RangeEnd inclusively. Unlike masked filters, the range filter does not work in MIXED mode. The user shall specify whether the range start, end or both are extended IDs by

manually setting the IDE bit (apply 0x40000000U mask over the desired ID).

The range filter type allows Standard Can ID type for Range Start and Extended Can ID type for Range End.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.139 Parameter RangeStart

Lower end of the range.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	0

4.140 Parameter RangeEnd

Higher end of the range.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4294967295
max	4294967295
min	0

4.141 Container CanAdvancedFeature

CanAdvancedFeature

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.142 Reference CanAdvancedFeatureRef

CanAdvancedFeatureRef

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Llce_Af/LlceAfGeneral/CanAdvancedFeature

4.143 Container CanTTHardwareObjectTrigger

This container is only included and valid if TTCAN SWS is used and TTCAN is enabled.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.144 Parameter CanTTHardwareObjectBaseCycle

Defines the cycle_offset.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	63
min	0

4.145 Parameter CanTTHardwareObjectCycleRepetition

Defines the repeat_factor.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	1
max	64
min	1

4.146 Parameter CanTTHardwareObjectTimeMark

Defines the point in time, when the trigger will be activated.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	65535
min	0

4.147 Parameter CanTTHardwareObjectTriggerId

Sequential number which allows separation of different TTCAN triggers configured for one and the same hardware object.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	63
min	0

4.148 Parameter CanTTHardwareObjectTriggerType

Defines the type of the trigger associated with the hardware object. This parameter depends on plain CAN parameter CAN_OBJECT_TYPE.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	CAN_TT_RX_TRIGGER
literals	['CAN_TT_RX_TRIGGER', 'CAN_TT_TX_REF_TRIGGER', 'CAN_TT_TX_REF_TRIGGER_GAP', 'CAN_TT_TX_TRIGGER_EXCLUSIVE', 'CAN_TT_TX_TRIGGER_MERGED', 'CAN_TT_TX_TRIGGER_SINGLE']

4.149 Container CanAuxFilter

Configuration for secondary lookup table

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.150 Parameter CanIdType

CAN065_Conf. Specifies whether the IdValue is of type

- standard identifier (ID - 11 bits length)
- extended identifier (ID - 29 bits length)
- mixed mode (standard or extended)

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	STANDARD
literals	['EXTENDED', 'MIXED', 'STANDARD']

4.151 Parameter Filter_Type

Type of filter

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	MASK
literals	['EXACT_MATCH', 'MASK', 'RANGE']

4.152 Parameter AuxSearchResult

Value to be returned on match

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	65535
min	0

4.153 Parameter CanHwFilterCode

CAN325_Conf. Specifies (together with the filter mask)- the identifiers range that passes the hardware filter for of RX objects.

Parameter ranges from 0 to 0x7FF (11 bits) for Standard IDs and 0 to 0xFFFFFFFF (29 bits) for Extended IDs.

User can assign any code to this parameter, but must to respect the above rule related to Standard/Extended IDs.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	0

4.154 Parameter CanHwFilterMask

ECUC_Can_00469 : Specifies (together with the filter mask) the identifiers range that passes the hardware filter.

Different CanHardwareObjects with different CanIdTypes (STANDARD, MIXED, EXTENDED)

can share the same CanFilterMask (i.e., the CanFilterMaskRef parameters of these

CanHardwareObjects reference the very same CanFilterMask container). This shall be allowed and must be supported by the configuration generators.

The CanFilterMaskRef is omitted for

- 1) CanHardwareObjects with CanObjectType set to TRANSMIT
- 2) CanHardwareObjects with CanObjectType set to RECEIVE if only a single Can ID

shall be received via this CanHardwareObjects(i.e. exact match with CanIdValue requested when CanHandleType is configured as FULL)

CAN321_Conf. Reference to the filter mask that is used for hardware filtering together with the CAN_ID_VALUE.

This value is used as acceptance masks for ID filtering in RX MBs.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

4.155 Parameter RangeStart

Lower end of the range.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	0

4.156 Parameter RangeEnd

High end of the range.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	4294967295
min	0

4.157 Reference CanControllerRef

CAN322_Conf. Reference to CAN Controller to which the HOH is associated to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Can/CanConfigSet/CanController

4.158 Container CanIcom

This container contains the parameters for configuring pretended networking

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Included subcontainers:

- [CanIcomConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.159 Container CanIcomConfig

This container contains the general configuration parameters of the ICOM Configuration

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Included subcontainers:

- [CanIcomWakeupCauses](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	256
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.160 Parameter CanIcomConfigId

This parameter identifies the ID of the ICOM configuration.

In order prevent the issue when have multiple configuration for ICom, Please configure the ConfigID follow the order.

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This field is not available

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1
max	255
min	1

4.161 Parameter CanIcomWakeOnBusOff

This parameter defines that the MCU shall wake if the bus off is detected or not.

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This field is not available

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.162 Container CanIcomWakeupCauses

This container contains the configuration parameters of the wakeup causes to leave the power saving mode.

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Included subcontainers:

- [CanIcomRxMessage](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.163 Container CanIcomRxMessage

This container contains the configuration parameters for the wakeup causes for matching received messages. It has to be configured as often as received messages are defined as wakeup cause. constraint: For all CanIcomRxMessage instances the Message IDs which are defined in CanIcomMessageId and in CanIcomRxMessageIdMask shall not overlap.

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Included subcontainers:

- [CanIcomRxMessageSignalConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.164 Parameter CanIcomCounterValue

This parameter defines that the MCU shall wake if the message with the ID is received n times on the communication channel.

NOTE: The ASR421 require 16 bit for this field, but hardware only support 8 bit for this field. So the limitation value of this field is 256.

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	1
max	255
min	1

4.165 Parameter CanIcomMessageId

This parameter defines the message ID the wakeup causes of this CanIcomRxMessage are configured for. In addition a mask (CanIcomMessageIdMask) can be defined, in that case it is possible to define a range of rx messages, which can create a wakeup condition.

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	536870912
min	0

4.166 Parameter CanIcomIdOperation

This is a non-autosar parameter. It is generated in order support for selection the ID filter type.

The Platform support 4 option in order ID filter wake-up message:

EXACTLY

SMALLER

GREATER

INSIDE_RANGE

PRETENDED NETWORKING IS NOT SUPPORTED ON THIS PLATFORM. This container is disabled

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	EXACTLY
literals	['EXACTLY', 'GREATER_MINNUM', 'SMALLER_MAXNUM', 'INSIDE_RANGE']

4.167 Parameter CanIcomMessageIdMask

Describes a mask for filtering of CAN identifiers. The CAN identifiers of incoming messages are masked with this CanIcomMessageIdMask. If the masked identifier matches the masked value of CanIcomMessageId, it can create a wakeup condition for this CanIcomRxMessage. Bits holding a 0 mean don't care, i.e. do not compare the message's identifier in the respective bit position. The mask shall be build by filling with leading 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	536870912
min	0

4.168 Parameter CanIcomMissingMessageTimerValue

This parameter defines that the MCU shall wake if the message with the ID is not received for a specific time in s on the communication channel.

NOTE: The '0' value have the meaning that the wake-up by timer disable, When you want to disable the wake-up by timer, you should disable this object.

The ASR421 don't require about the limitation value for this field, but the hardware only support 16 bit for this field, so this field has the 0-65535 range as the limitation range.

The internal timer is incremented based on periodic time ticks, which period is 64 times the CAN Bit Time unit.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0.0
max	65535.0
min	0.0

4.169 Parameter CanIcomPayloadLengthError

This parameter defines that the MCU shall wake if a payload error occurs

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.170 Container CanIcomRxMessageSignalConfig

This container contains the configuration parameters for the wakeup causes for matching signals.

It has to be configured as often as a signal is defined as wakeup cause. If at least one Signal conditions defined in a CanIcomRxMessageSignalConfig evaluates to true or if no CanIcomRxMessageSignalConfig are defined, the whole wakeup condition is considered to be true. All instances of this container refer to the same frame/pdu (see CanIcomMessageId).

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.171 Parameter CanIcomSignalMask

This parameter shall be used to mask a signal in the payload of a CAN message. The mask is binary AND with the signal payload. The result will be used in combination of the operations defined in CanIcomSignalOperation with the CanIcomSignalValue.

the ASR request for full 64 bit with Integers type. but in the Tresos tool, the Integers only has 63 bit, So in the fact, the greatest value is 0x7fffffffffffff.

Property	Value
type	ECUC-INTEGGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	0
max	9223372036854775807
min	0

4.172 Parameter CanIcomSignalOperation

This parameter defines the operation, which shall be used to verify the signal value creates a wakeup condition.

The Calypso platform don't support the "AND" filter type.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	EQUAL
literals	['AND', 'EQUAL', 'GREATER', 'SMALLER', 'XOR']

4.173 Parameter CanIcomSignalValue

This parameter shall be used to define a signal value which shall be compared (CanIcomSignalOperation) with the masked CanIcomSignalMask value of the received signal (CanIcomSignalRef).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	9223372036854775807
min	0

4.174 Parameter DLCLowValue

This is a non-autosar object. It is used to configure the lowest value for the "CAN_FLT_DLC" register.

That value is number data byte lowest of messages wake-up.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	8
min	0

4.175 Parameter DLCHighValue

This is a non-autosar object. It is used to configure the highest value for the "CAN_FLT_DLC" register.

That value is number data byte highest of messages wake-up.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0
max	8
min	0

4.176 Reference CanIcomSignalRef

This parameter defines a reference to the signal which shall be checked additional to the message id (CanIcomMessageId). This reference is used for documentation to define which ComSignal originates this filter setting. All signals being referred by this reference shall point to the same PDU.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Com/ComConfig/ComSignal

4.177 Container CommonPublishedInformation

Common container, aggregated by all modules.

It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.178 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.179 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.180 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.181 Parameter ModuleId

Module ID of this module from Module List.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	80
max	80
min	80

4.182 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	1
max	1
min	1

4.183 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.184 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	7
max	7
min	7

4.185 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the Implementation specific name is generated as follows:

<ModuleName>_>VendorId>_<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name

Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity >

1. It shall not be used for modules with upper multiplicity =1.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	LLCE

4.186 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

This chapter describes the Tresos configuration plug-in for the CAN_43_LLCE Driver. The most of the parameters are described below.

How to Reach Us:

Home Page:

nxp.com

Web Support:

nxp.com/support

Information in this document is provided solely to enable system and software implementers to use NXP products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits based on the information in this document. NXP reserves the right to make changes without further notice to any products herein.

NXP makes no warranty, representation, or guarantee regarding the suitability of its products for any particular purpose, nor does NXP assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in NXP data sheets and/or specifications can and do vary in different applications, and actual performance may vary over time. All operating parameters, including "typicals," must be validated for each customer application by customer's technical experts. NXP does not convey any license under its patent rights nor the rights of others. NXP sells products pursuant to standard terms and conditions of sale, which can be found at the following address: nxp.com/SalesTermsandConditions.

NXP, the NXP logo, NXP SECURE CONNECTIONS FOR A SMARTER WORLD, COOLFLUX, EMBRACE, GREENCHIP, HITAG, I2C BUS, ICODE, JCOP, LIFE VIBES, MIFARE, MIFARE CLASSIC, MIFARE DESFire, MIFARE PLUS, MIFARE FLEX, MANTIS, MIFARE ULTRALIGHT, MIFARE4MOBILE, MIGLO, NTAG, ROADLINK, SMARTLX, SMARTMX, STARPLUG, TOPFET, TRENCHMOS, UCODE, Freescale, the Freescale logo, Altivec, C-5, CodeTEST, CodeWarrior, ColdFire, ColdFire+, C-Ware, the Energy Efficient Solutions logo, Kinetis, Layerscape, MagniV, mobileGT, PEG, PowerQUICC, Processor Expert, QorIQ, QorIQ Qonverge, Ready Play, SafeAssure, the SafeAssure logo, StarCore, Symphony, VortiQa, Vybrid, Airfast, BeeKit, BeeStack, CoreNet, Flexis, MXC, Platform in a Package, QUICC Engine, SMARTMOS, Tower, TurboLink, and UMEMS are trademarks of NXP B.V. All other product or service names are the property of their respective owners. ARM, AMBA, ARM Powered, Artisan, Cortex, Jazelle, Keil, SecurCore, Thumb, TrustZone, and Vision are registered trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. ARM7, ARM9, ARM11, big.LITTLE, CoreLink, CoreSight, DesignStart, Mali, mbed, NEON, POP, Sensinode, Socrates, ULINK and Versatile are trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. All rights reserved. Oracle and Java are registered trademarks of Oracle and/or its affiliates. The Power Architecture and Power.org word marks and the Power and Power.org logos and related marks are trademarks and service marks licensed by Power.org.

© 2023 NXP B.V.

